

Rahul Verma

rahul.verma@berkeley.edu | [linkedin.com/in/rahul-verma-berkeley](https://www.linkedin.com/in/rahul-verma-berkeley)

EDUCATION

University of California, Berkeley	Berkeley, CA
<i>B.S. Electrical Engineering and Computer Sciences, Minor in Mechanical Engineering</i>	<i>Aug. 2026 – May 2029</i>
– Current Coursework: EE 64 (Circuits), EE 66 (Signals, Dynamics, & Information), CS 61A (Python)	
– Prior Coursework: Multivariable Calculus, Linear Algebra, Differential Equations — MiraCosta College	
Del Norte High School	San Diego, CA
<i>GPA: 4.46 Weighted; ACT: 36</i>	<i>2022 – 2026</i>

EXPERIENCE

UCSD Supercomputer Center <i>Research Intern</i>	Summer 2024, 2025
– Benchmarked & optimized ML models in PyTorch/TensorFlow across NVIDIA A100/V100 GPUs, finding superior A100 performance on medical/image-generation models and diminishing efficiency returns past 32 CPU cores	
– Optimized supercomputer usage for real-time agricultural monitoring and detecting of soybean fields in Iowa	
– Presented research poster on the newly designed benchmarking workflow; poster featured on ICICLE research organization website and workflow now used across center	
FIRST Tech Challenge – Team 22105 <i>Founder, Co-Captain, CAD/Design Lead</i>	2022 – 2026
– Founded and led a 15-student team across five schools; designed, built, and programmed seven competition robots	
– Integrated mechanical, electrical, & software systems, leading to qualifying for the FTC World Championship	
Competitive Programming Initiative <i>Project Manager for National Nonprofit Organization</i>	2023 – 2025
– Coordinated 8 ten-week computer science courses for 250+ students to prepare for the USA Computing Olympiad	
– Created/taught lessons in competitive programming, managed/onboarded instructors, and handled finances	

PROJECTS

Omni-Based Swerve Drive <i>CAD, Kinematics, Control Theory</i>	2024 – 2025
– Designed and built a coaxial swerve-omni drivetrain that increases forward speed past the motor's nominal limit by tilting each wheel's yaw axis, achieving an effect similar to a continuously variable transmission (CVT)	
– Developed a velocity-based nonlinear control algorithm, balancing speed against traction loss. Achieved a 1.63x speed boost and won 1st Place in Engineering: Robotics at Greater San Diego Science and Engineering Fair	
Telescopic and Pivoting Arm <i>CAD, Controls, Mechanism Design</i>	2025 – 2026
– Engineered a telescoping, pivoting aluminum-tube arm (4 nested pocketed tubes, 2" to 0.5") through 48 design iterations, achieving 4x rigidity and 47% less weight than commercial linear slides with under 5mm of backlash	
– Derived a gain-scheduling feedforward controller and custom Newton-Raphson S-curve motion profile to account for the arm's variable moment of inertia, cutting full extension to 0.4s and 90° pivots to 0.25s	

LEADERSHIP

Del Norte Algorithmic Coding Club <i>President</i>	2023 – 2026
– Led club of 30+ to 15th nationwide in the American Computer Science League; taught competitive programming, boolean algebra, digital electronics, Regex operations, LISP, and Assembly	

AWARDS

FTC Michiana International Premier Event – 3rd Place overall across 96 teams	2025
American Computer Science League – Silver Medalist, Senior Division, top 20 of 2000+	2025
National Merit Finalist – Top 1% of U.S. HS Students on PSAT Exam	2026
AIME Qualifier 2x – Top 5% scorer on American Mathematics Competition examination	2022, 2023

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL
Engineering: Mechanical Design, Robotics, PCB Design, 3D Printing, CNC Machining, Systems Integration
CAD: SolidWorks, Onshape, Fusion360, Altium Designer
ML / Computing: PyTorch, TensorFlow, NumPy, Matplotlib, Linux, SLURM